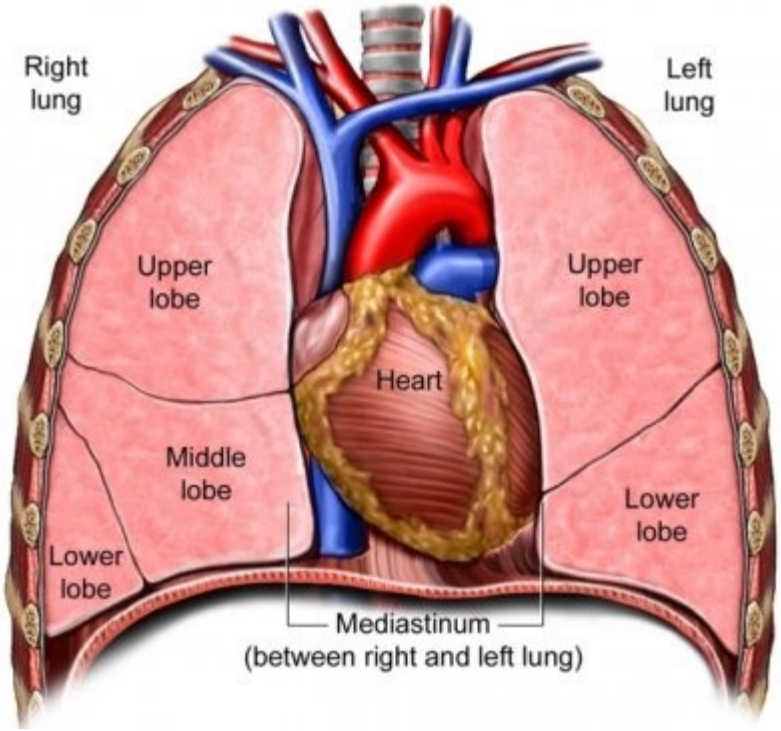
MEDIASTINUM

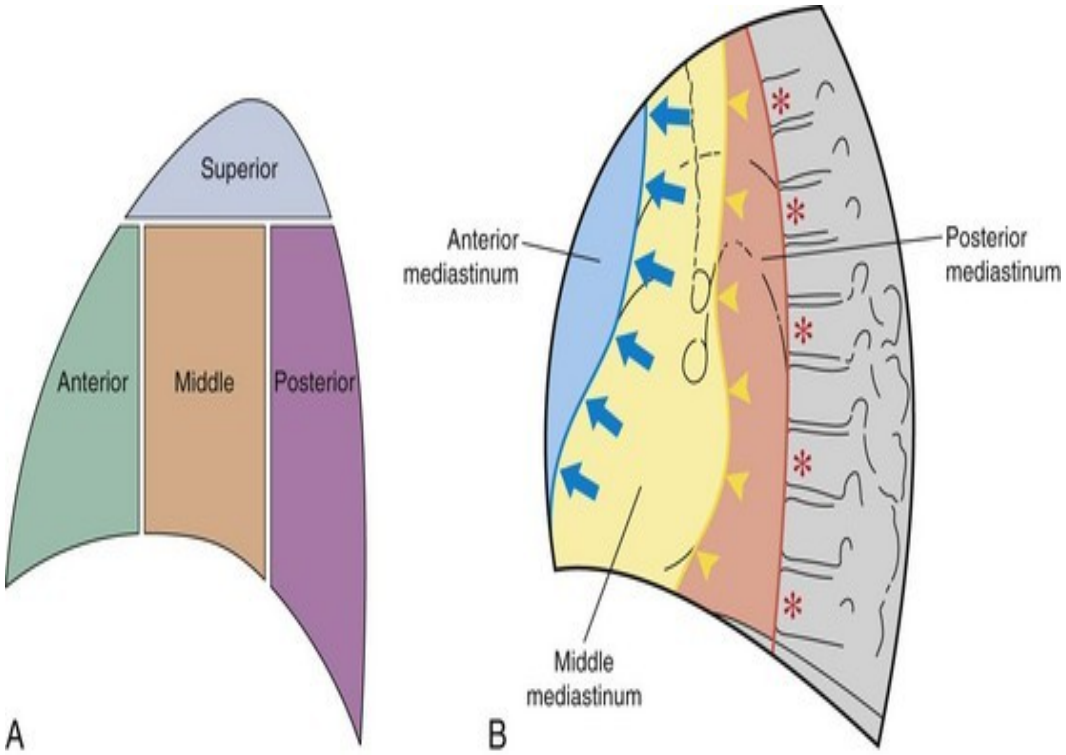
Features

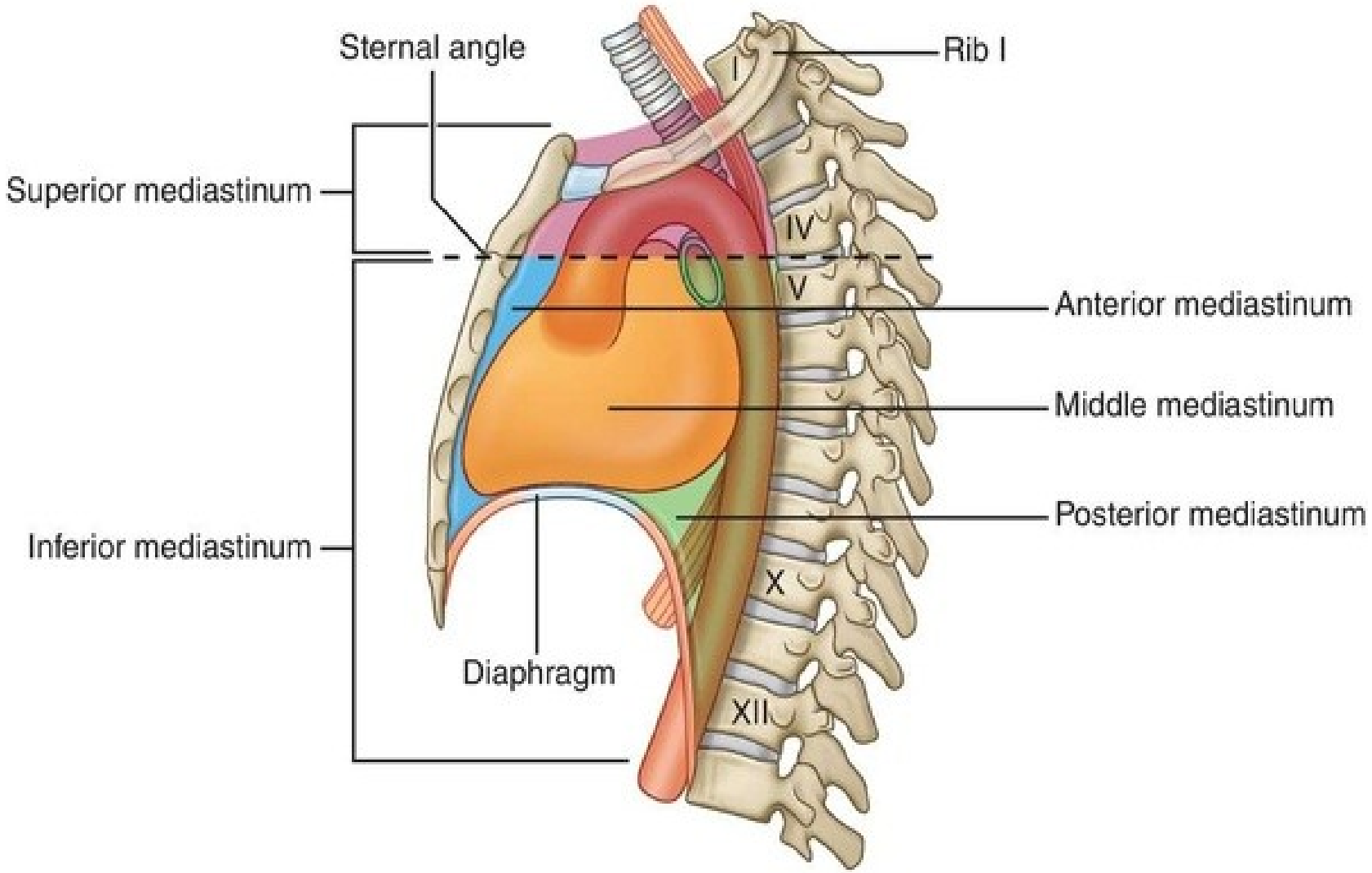
- Central, midline compartment of thoracic cavity
- Boundaries:
- anteriorly : by sternum
- Posteriorly: by 12 thoracic vertebrae
- laterally : by pleural cavities
- Superiorly: thoracic inlet
- Inferiorly: diaphragm



Divisions

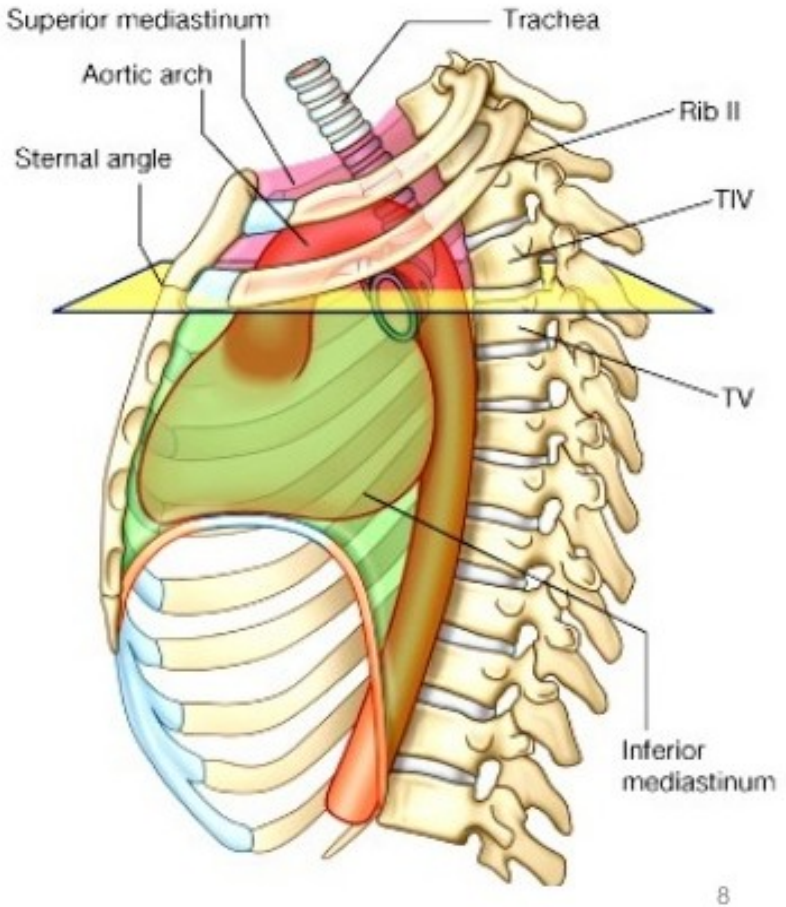
- Mediastinum is divided into superior and inferior mediastina by a horizontal plane.
- Plane: from sternal angle anteriorly to IV Disc T4/5 posteriorly
- Inferior mediastinum is further divided into anterior , middle and posterior mediastina.




Sternal angle
Rib I
I
Superior mediastinum
IV
V
Anterior mediastinum
Middle mediastinum
Posterior mediastinum
Inferior mediastinum
X
Diaphragm
XII

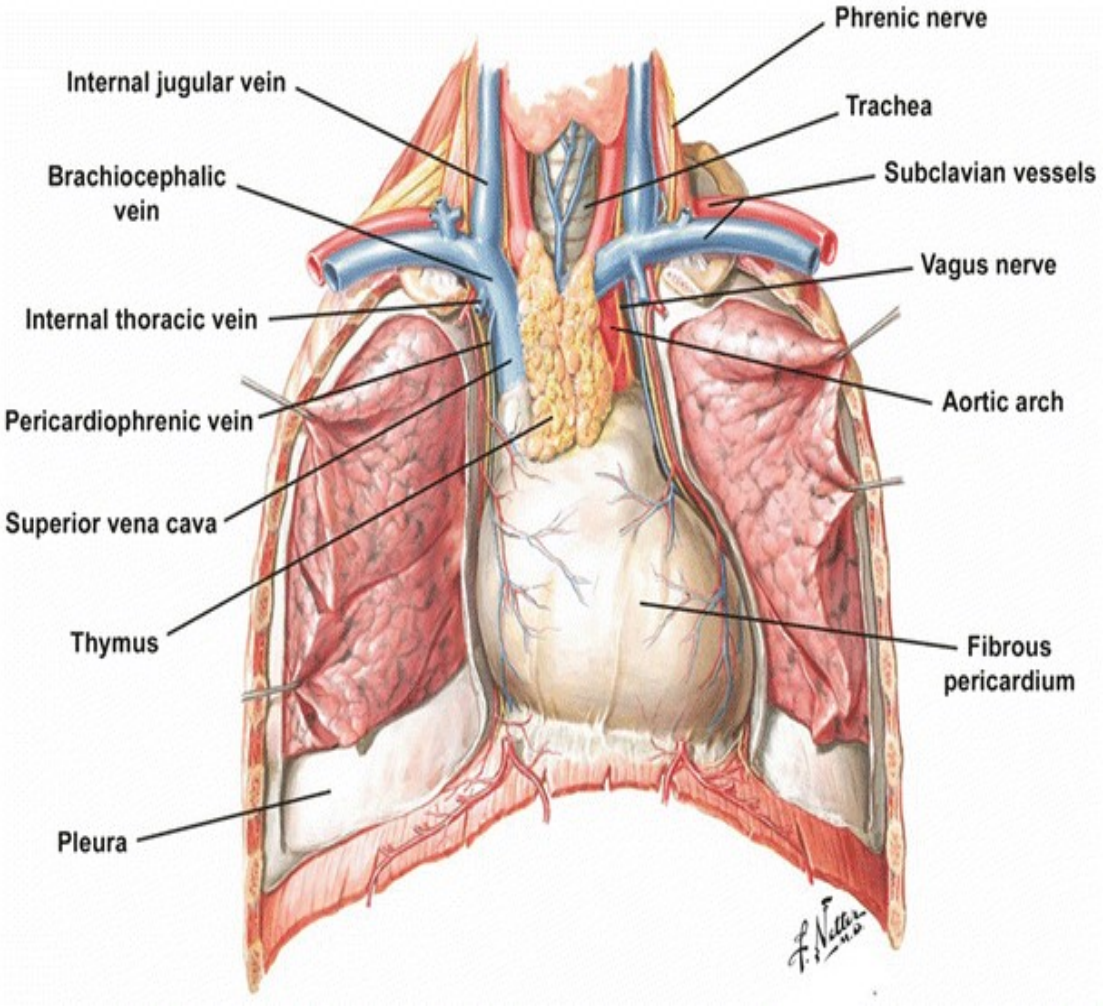
Superior Mediastinum

- Anteriorly: Manubrium of sternum
- Posteriorly: vertebral bodies of T1-T4
- Laterally: Lungs and pleura
- Superiorly: Thoracic inlet
- Inferiorly: inferior mediastenum



Anterior Mediastinum

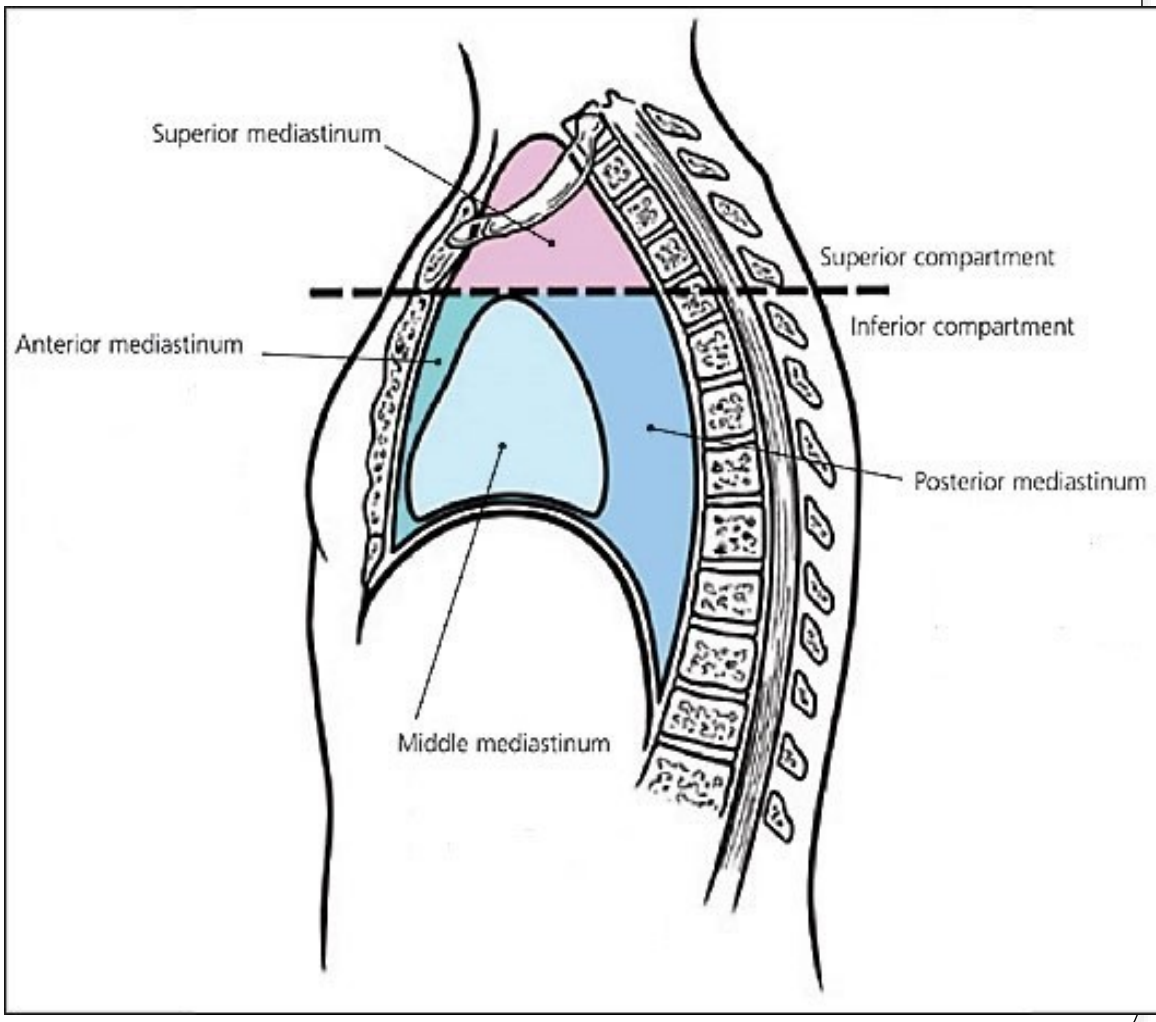
- Small interval b/w sternum and anterior surface of pericardium.
- Anteriorly: body of the sternum.
- Posteriorly: pericardium
- Laterally: mediastenal pleura
- Roof: superior mediastenum (at sternal angle).
- Floor: diaphragm



Middle Mediastinum

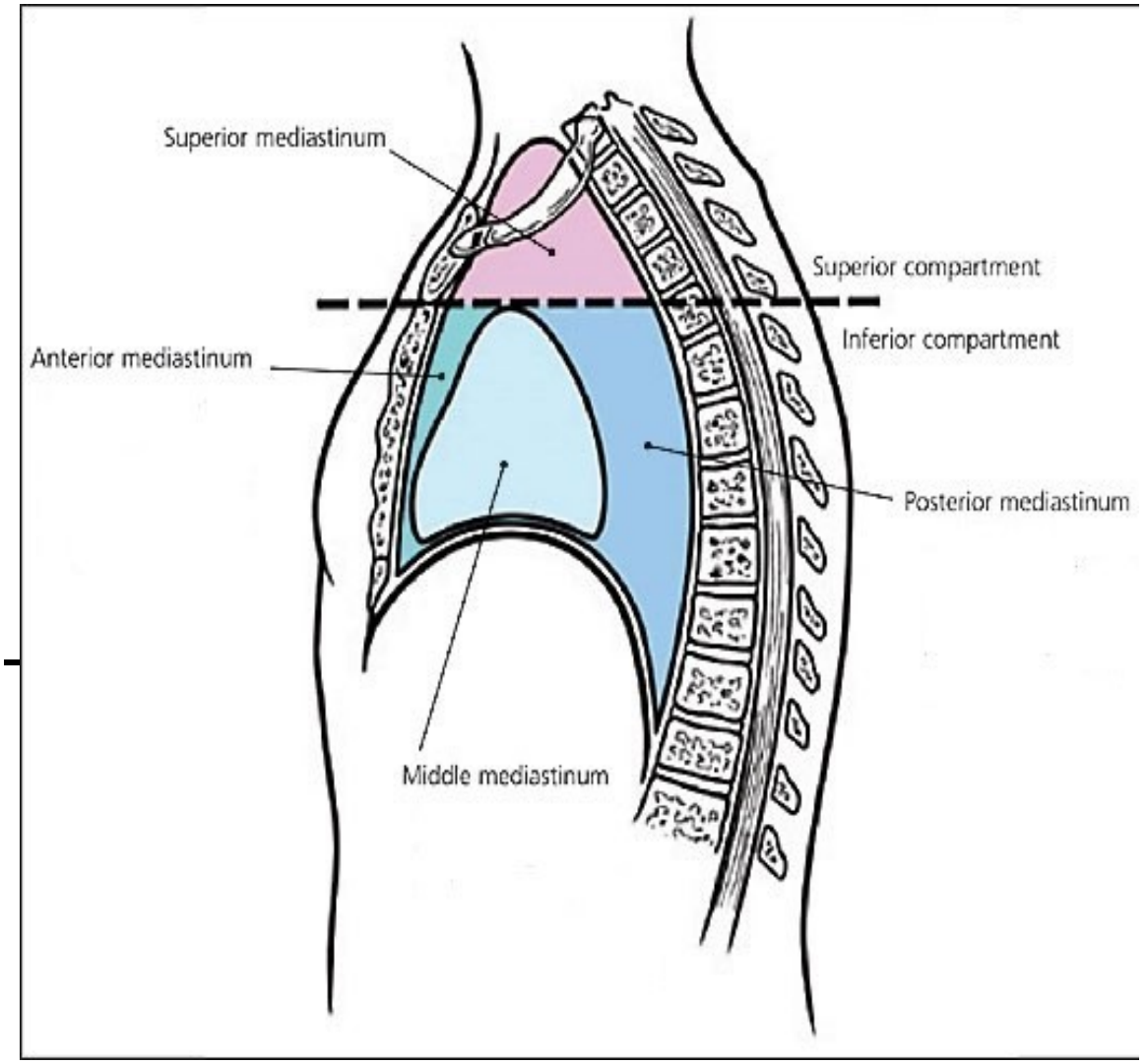
Boundaries:

- Anteriorly: anterior margin of pericardium.
- Posteriorly: posterior border of pericardium
- Laterally: mediastenal pleura of lungs
- Superiorly: sternal angle-t4,t5
- Inferiorly: diaphragm



Posterior Mediastinum

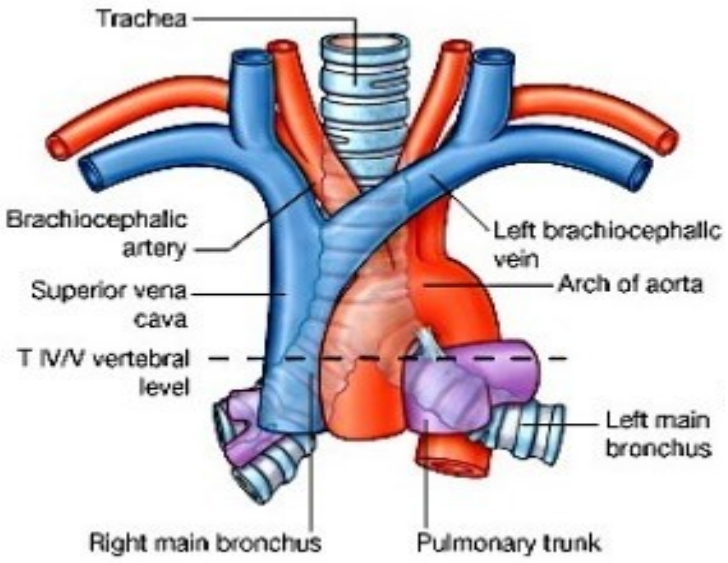
- Anterior: pericardium
- Posteriorly: T5-T12 vertebrae
- Laterally: mediastenal pleura
- Roof: : sternal angle-t4,t5
- Floor: diaphragm

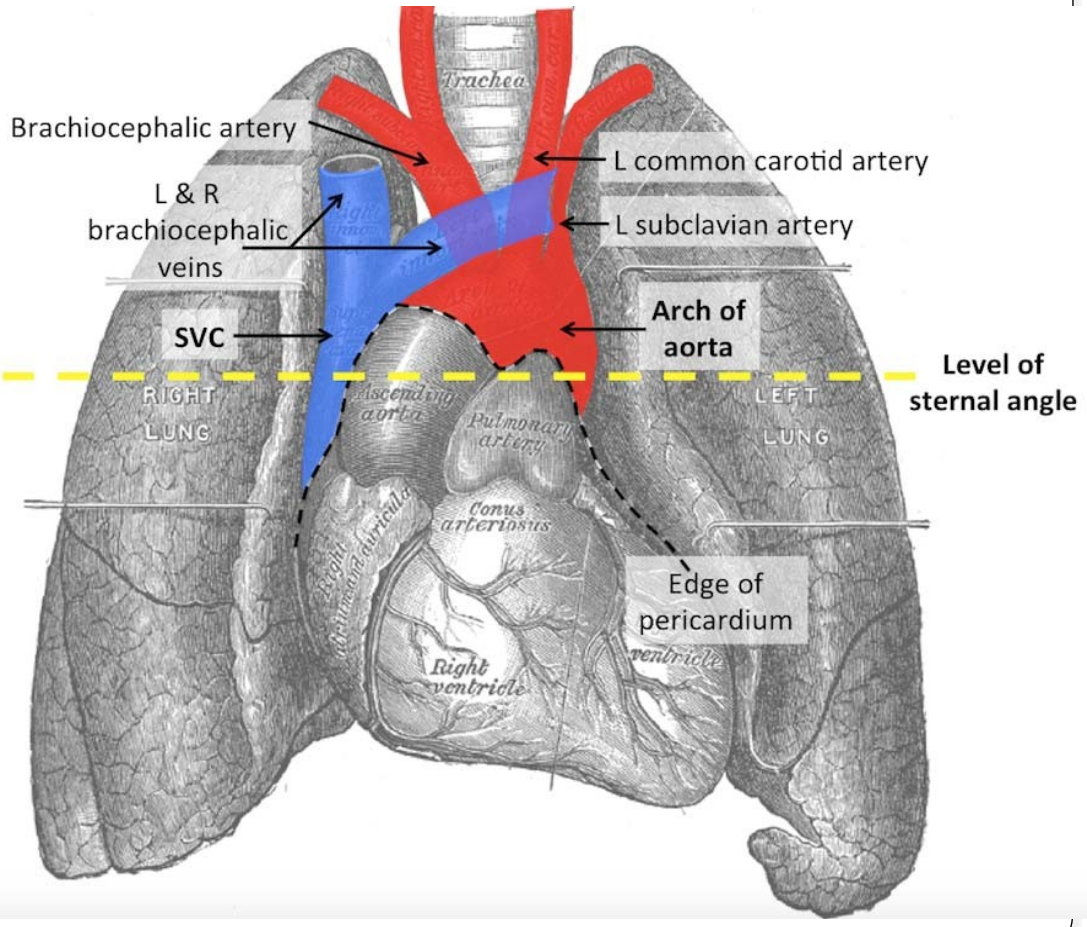


Contents of mediastenum

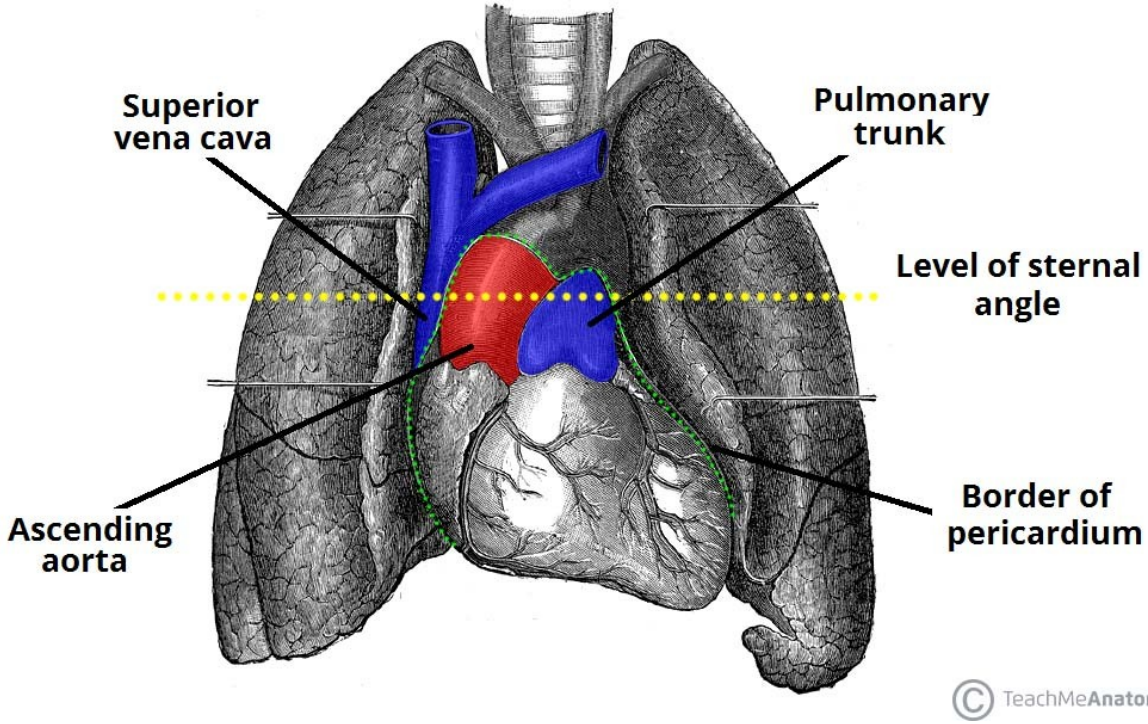
SUPERIOR	ANTERIOR	MIDDLE	POSTERIOR
•Thymus •Trachea(and its bifurcation) •Esophagus •Arch of aorta(with 3 branches) •Superior vana cava •Brachiocephalic veins (lf & rt) •Thoracic duct •Phrenic nerve •Vagus nerve •Left recurrent laryngeal nerve	•Remnants of thymus •lymphnodes	•Heart •Pericardium (parietal and fibrous) •Roots of great vessels: ➢ Aorta ➢ Pulmonary trunk ➢ Pulmonary arteries ➢ Pulmonary veins ➢ Vana-cavae •Main bronchi (left & right) •Phrenic nerves •Arch of azygous vein	•Esophagus •Thoracic aorta •Thoracic duct •Azygous vein •Hemi-azygous vein •Vagus nerve •Splanchnic nerves •Sympathetic trunk

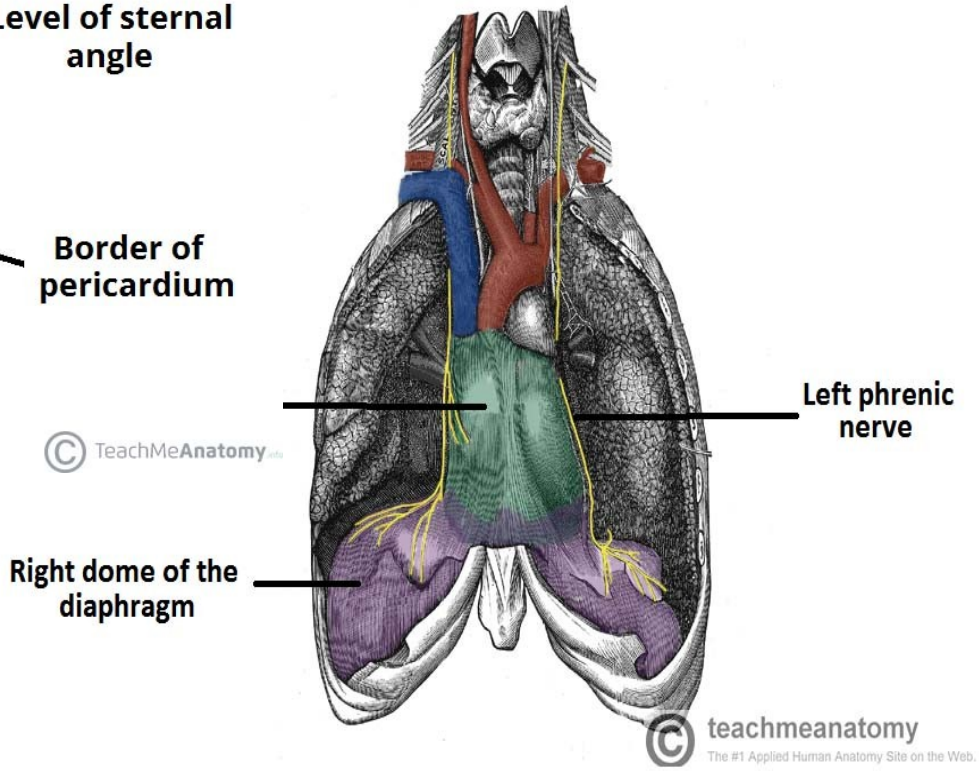
Superior mediastenum



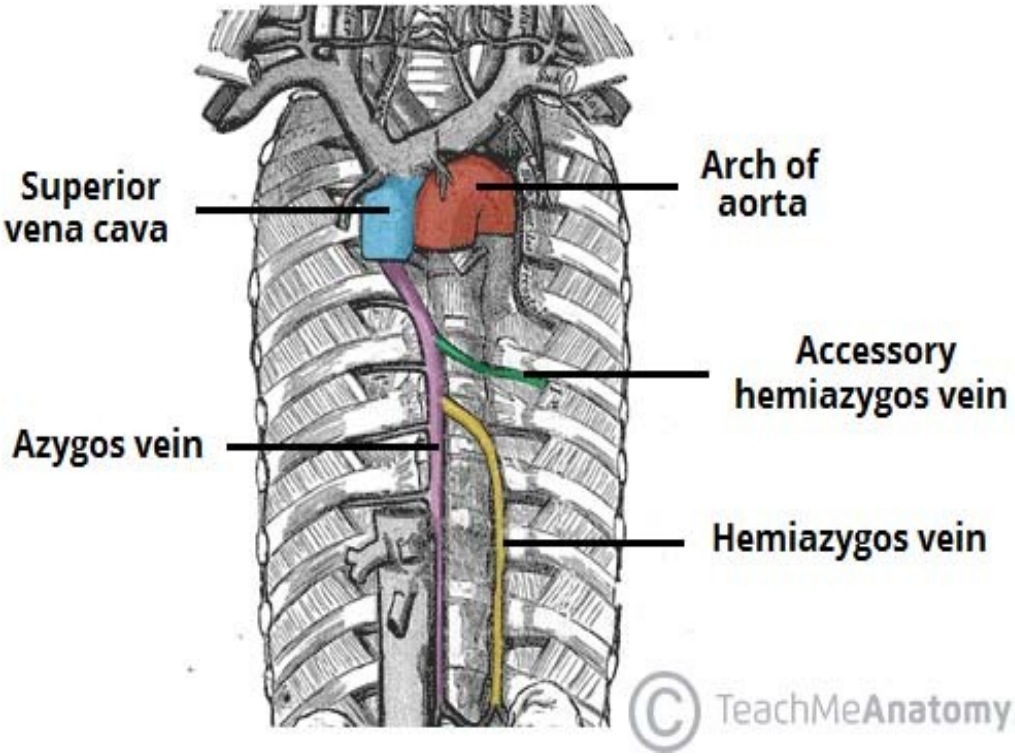


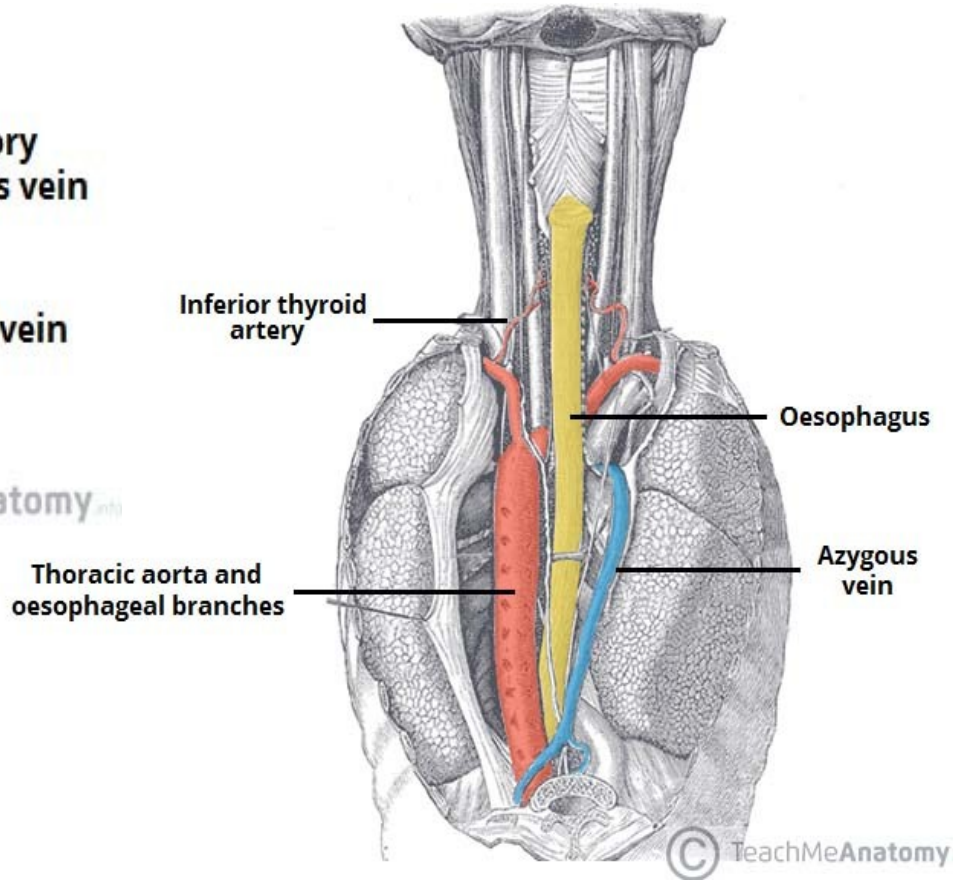
Middle mediastenum





Posterior mediastenum





Pleura, Lungs, Trachea

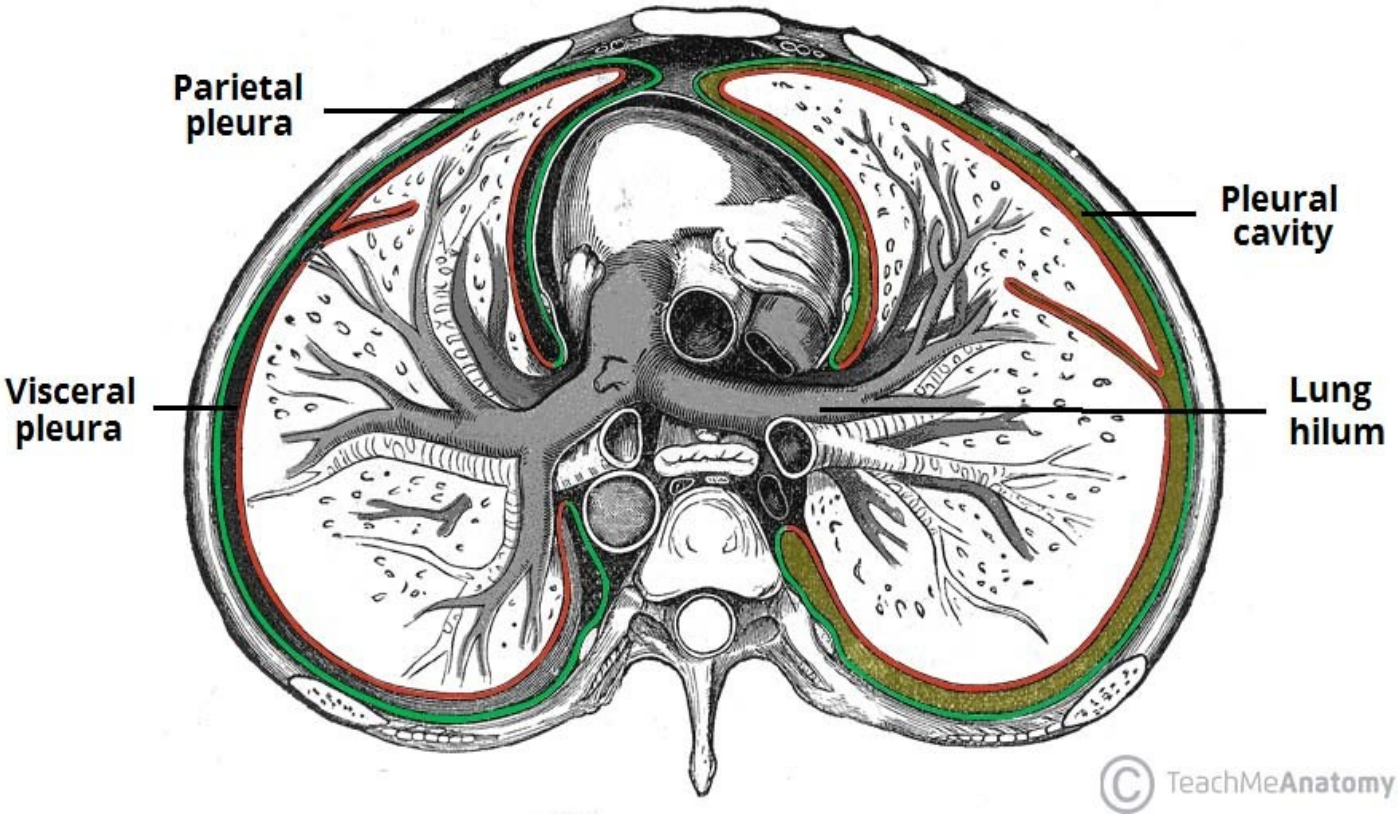
Pleura

- The pleurae refer to the **serous membranes** that line the lungs and thoracic cavity. They permit efficient and effortless respiration.

- Each pleura can be divided into two parts:
  - **Visceral pleura** – covers the lungs.
  - **Parietal pleura** – covers the internal surface of the thoracic cavity.
  - There is a potential space between the viscera and parietal pleura, known as the pleural cavity.
- These two parts are continuous with each other at the **hilum** of each lung.

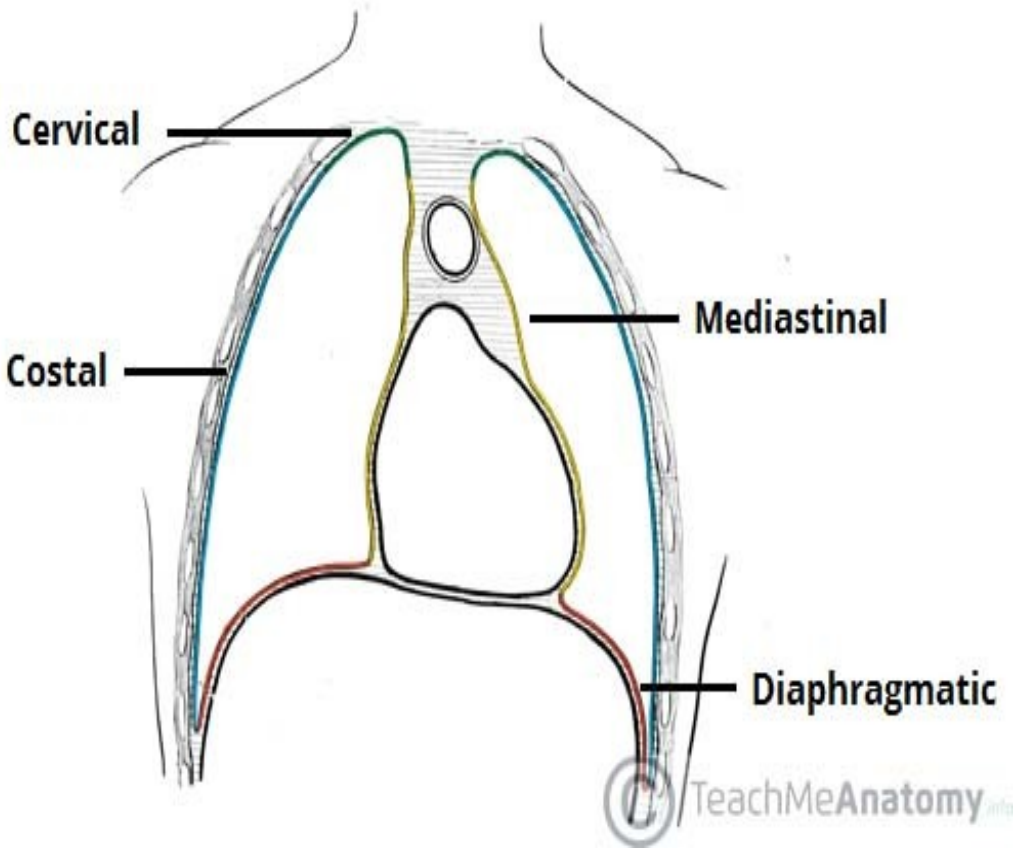
Visceral Pleura

- The visceral pleura covers the outer surface of the lungs, and extends into the interlobar fissures. It is continuous with the parietal pleura at the **hilum** of each lung



Parietal Pleura

- The parietal pleura covers the internal surface of the thoracic cavity. It is thicker than the viscera pleura, and can be subdivided according to the part of the body that it is contact with:
- **Mediastinal pleura** – Covers the lateral aspect of the mediastinum (the central component of the thoracic cavity, containing a number of organ).
- **Cervical pleura** – Lines the extension of the pleural cavity into the neck.
- **Costal pleura** – Covers the inner aspect of the ribs, costal cartilages, and intercostal muscles.
- **Diaphragmatic pleura** – Covers the thoracic (superior) surface of the diaphragm.

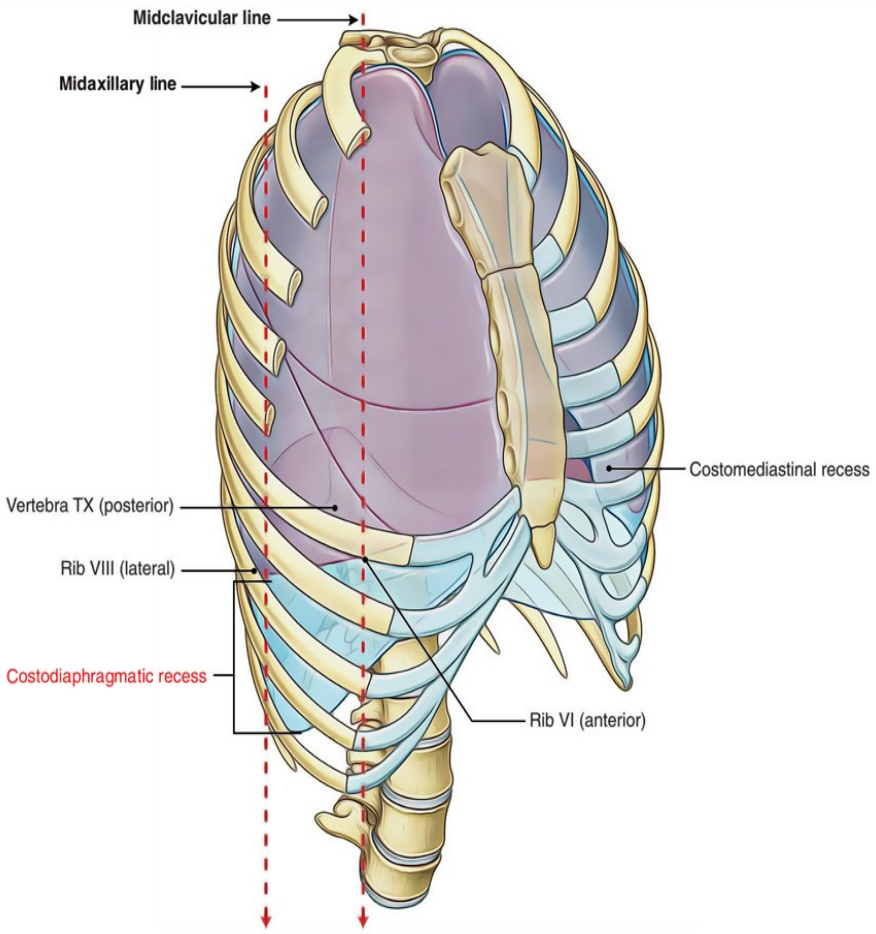


Pleural Cavity

- The pleural cavity is a **potential space** between the parietal and visceral pleura. It contains a small volume of serous fluid, which has two major functions.
- It lubricates the surfaces of the pleurae, allowing them to slide over each other.
- The serous fluid also produces a surface tension, pulling the parietal and visceral pleura together. This ensures that when the thorax expands, the lung also expands, filling with air.

Pleural Recesses

- Anteriorly and posteroinferiorly, the pleural cavity is not completely filled by the lungs. This gives rise to **recesses** – where the opposing surfaces of the parietal pleura touch.
- There are two recesses present in each pleural cavity:
- **Costodiaphragmatic** – located between the costal pleurae and the diaphragmatic pleura.
- **Costomediastinal** – located between the costal pleurae and the mediastinal pleurae, behind the sternum.
- These recesses are of clinical importance, as they provide a location where fluid can collect (such as in a pleural effusion).



Neurovascular Supply

- The two parts of the pleurae receive a different neurovascular supply:
- **Parietal Pleura**
- The parietal pleura is sensitive to pressure, pain, and temperature. It produces a well localised pain, and is innervated by the **phrenic** and **intercostal nerves**.
- The blood supply is derived from the **intercostal arteries**.
- **Visceral Pleura**
- The visceral pleura is not sensitive to pain, temperature or touch. Its sensory fibres only detect stretch. It also receives autonomic innervation from the **pulmonary plexus** (a network of nerves derived from the sympathetic trunk and vagus nerve).
- Arterial supply is via the **bronchial arteries** (branches of the descending aorta), which also supply the parenchyma of the lungs

Clinical Relevance: Pneumothorax

- A **pneumothorax** (commonly referred to a collapsed lung) occurs when air or gas is present within the pleural space.
- Clinical features include chest pain, shortness of breath and asymmetrical chest expansion.
- **Spontaneous:** A spontaneous pneumothorax occurs without a specific cause.
  >>primary (no underlying respiratory disease)
  >> secondary (underlying respiratory disease present).
- **Traumatic:** A traumatic pneumothorax occurs as a result of blunt or penetrating chest trauma, such as a rib fracture (in road traffic accident).
- Treatment: thoracostomy/thoracotomy.

Lungs

- The lungs are roughly **cone** shaped, Separated from each other by mediastinum
- The left lung is slightly smaller than the right – this is due to the presence of the heart.
- Each lung consists of an apex, base, lobes, fissure, surfaces and borders.

  - **Apex** – The blunt superior end of the lung. It projects upwards, above the level of the 1st rib and into the floor of the neck.

  - **Base** : the inferior diaphragmatic surface of lung which sits on the diaphragm.

  - **Lobes** (two or three) – These are separated by fissures within the lung.

>>**Oblique fissures-** of both lungs project anteriorly in 5th ICS in mid clavicular line ending medially deep to 6th CC

>>**Horizontal fissure**– Runs horizontally from the sternum, at the level of the 4th rib, to meet the oblique fissure.

- **Surfaces** (three) – These correspond to the area of the thorax that they face. They are:

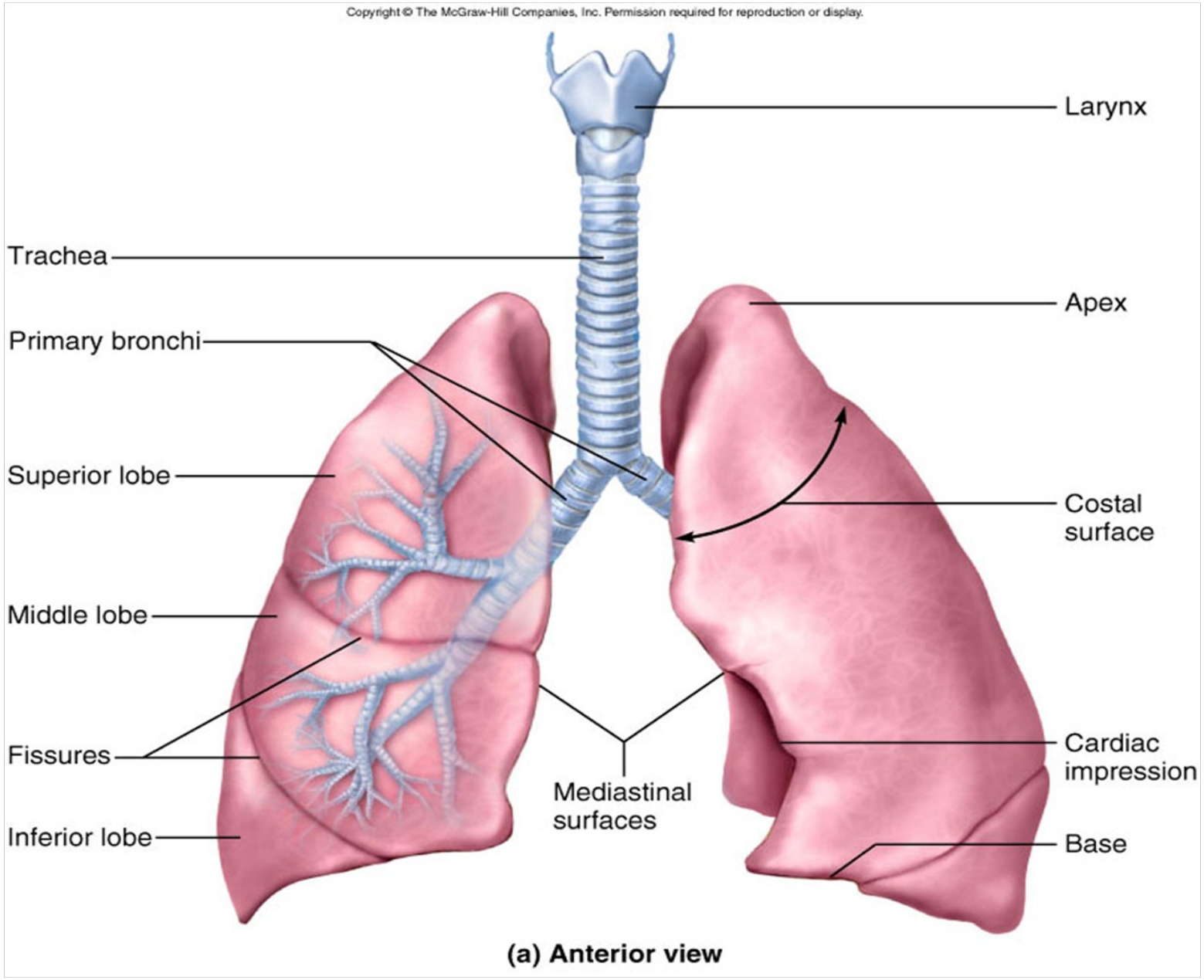
>Costal: smooth and convex, laterally related to ribs and tissues of chest wall

>Mediastinal: concave and medially relates to middle mediastinum and heart. This surface contains root of lung and deep cardiac impression

>Diaphragmatic: – The concave inferior surface of the lung that rests on superior surface of diaphragm, more sup: on right due to liver.

- **Borders** (three) –
- Anterior: formed by the convergence of the mediastinal and costal surfaces
- Inferior: separates the base of the lung from the costal and mediastinal surfaces
- Posterior borders: is smooth and rounded. It is formed by the costal and mediastinal surfaces meeting posteriorly.

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(a) Anterior view

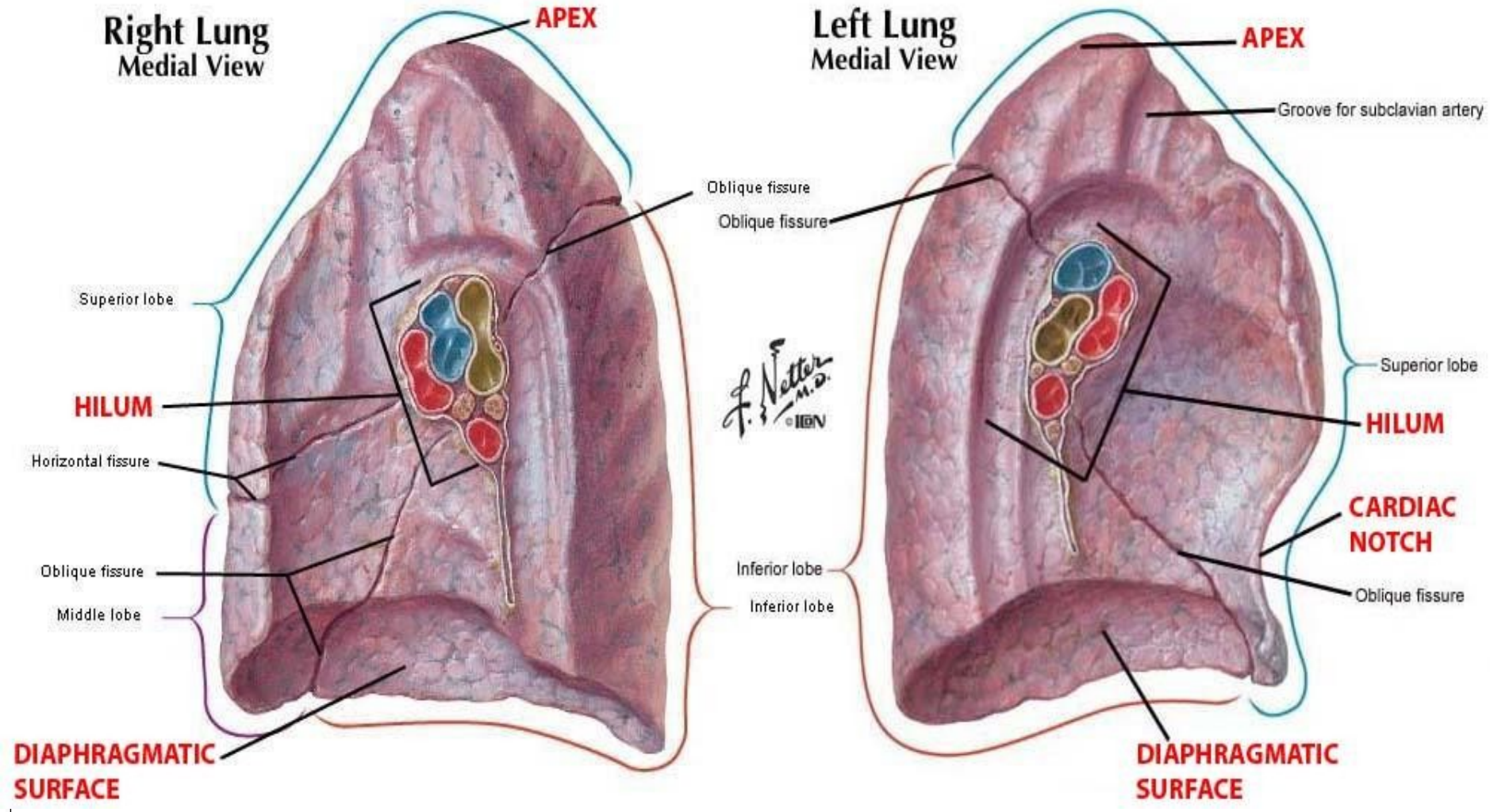
Right Lung

- Larger and heavier
- Shorter and wider because of high right dome of diaphragm.
- 3 lobes, 3 lobar (secondary bronchi)
- 2 fissures:
- 10 segmental (tertiary) bronchi
- Grooves for various structures i.e SVC, IVC, azygous vein, oesophagus
- Receive single bronchial artery

Left lung

- 2 lobes, 2 lobar bronchi
- 8-10 segmental bronchi
- Contain cardiac notch at ant border of superior lobe.
- Grooved for aortic arch, thoracic aorta, left sub clavian artery
- Recieves two bronchial arteries

- **Root and Hilum**
- The lung **root** is a collection of structures that suspends the lung from the mediastinum. Each root contains a bronchus, pulmonary artery, two pulmonary veins, bronchial vessels, pulmonary plexus of nerves and lymphatic vessels.
- All these structures enter or leave the lung via the **hilum** – a wedge shaped area on its mediastinal surface.


Right Lung
Medial View
APEX
Oblique fissure
Superior lobe
HILUM
Horizontal fissure
Oblique fissure
Middle lobe
DIAPHRAGMATIC SURFACE
Inferior lobe
Left Lung
Medial View
APEX
Groove for subclavian artery
Oblique fissure
Superior lobe
HILUM
CARDIAC NOTCH
Inferior lobe
Oblique fissure
DIAPHRAGMATIC SURFACE

Broncho-pulmonary segments of lungs

- It is Anatomical, functional and surgical unit.
- Each segment Consists of:
  - Segmental bronchus
  - segmental branch of pulmonary artery
  - segment of lung tissue(surrounded by intersegmental septum)
  - intersegmental pulmonary veins.

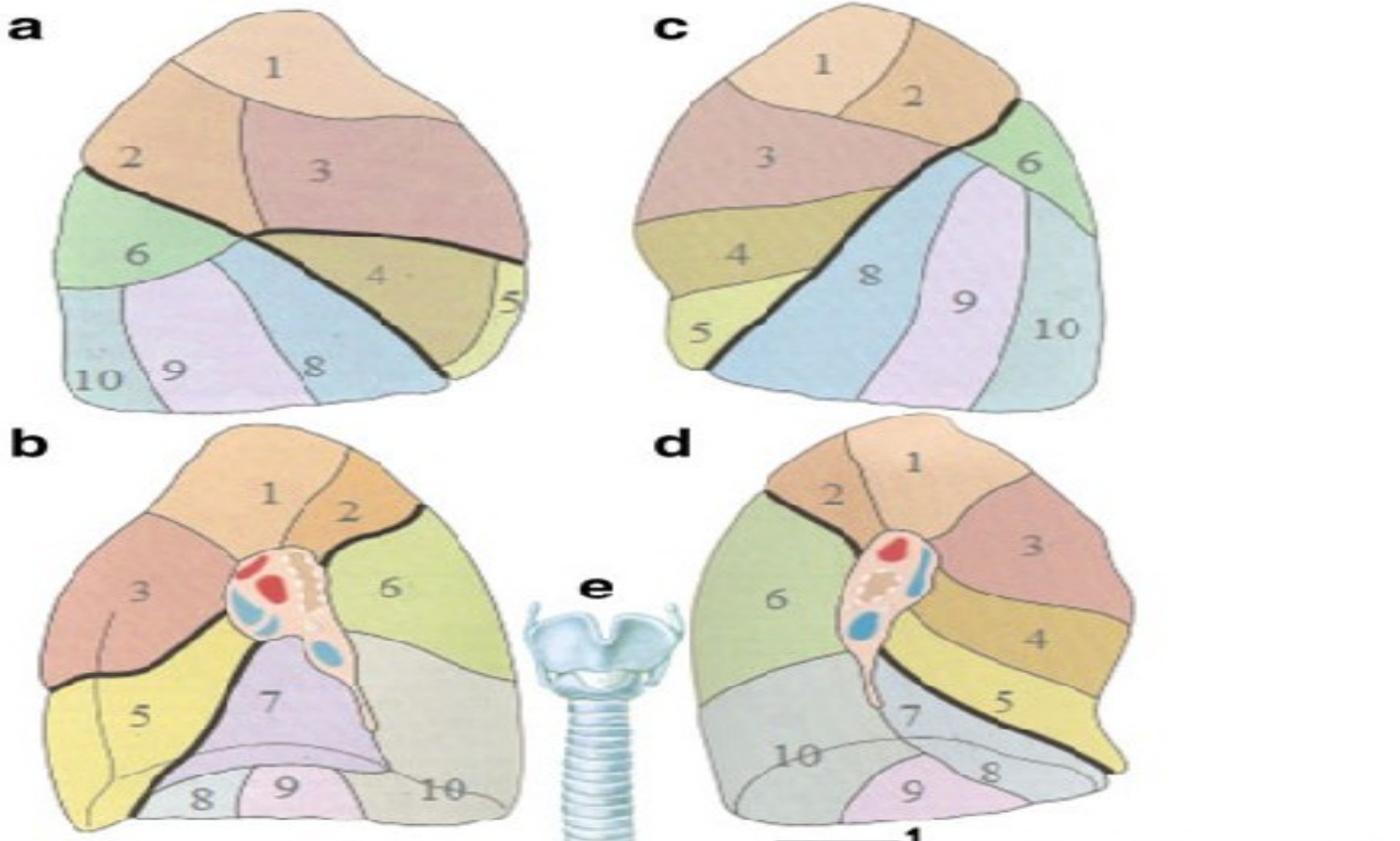
Bronchopulmonary segments

Right lobe

- Upper lobe:
  apical, posterior, anterior
- Middle lobe:
  medial, lateral
- Lower lobe:
  superior, media basal, lateral basal, anterior basal, posterior

Left lobe

- Upper lobe:
  apico posterior, anterior, inferior lingular, superior lingular
- Lower lobe:
  superior, anterio medial, posterio basal, lateral basal



Right lung (Upper lobe):
1. Apical
2. Posterior
3. Anterior

Right lung (Middle lobe):
4. Lateral
5. Medial

Right lung (Lower lobe):
6. Superior
7. Medial basal
8. Anterior basal
9. Lateral basal
10. Posterior basal

Left lung (Upper lobe):
1. Apical
2. Posterior
3. Anterior
4. Lingula-Superior
5. Lingula Inferior

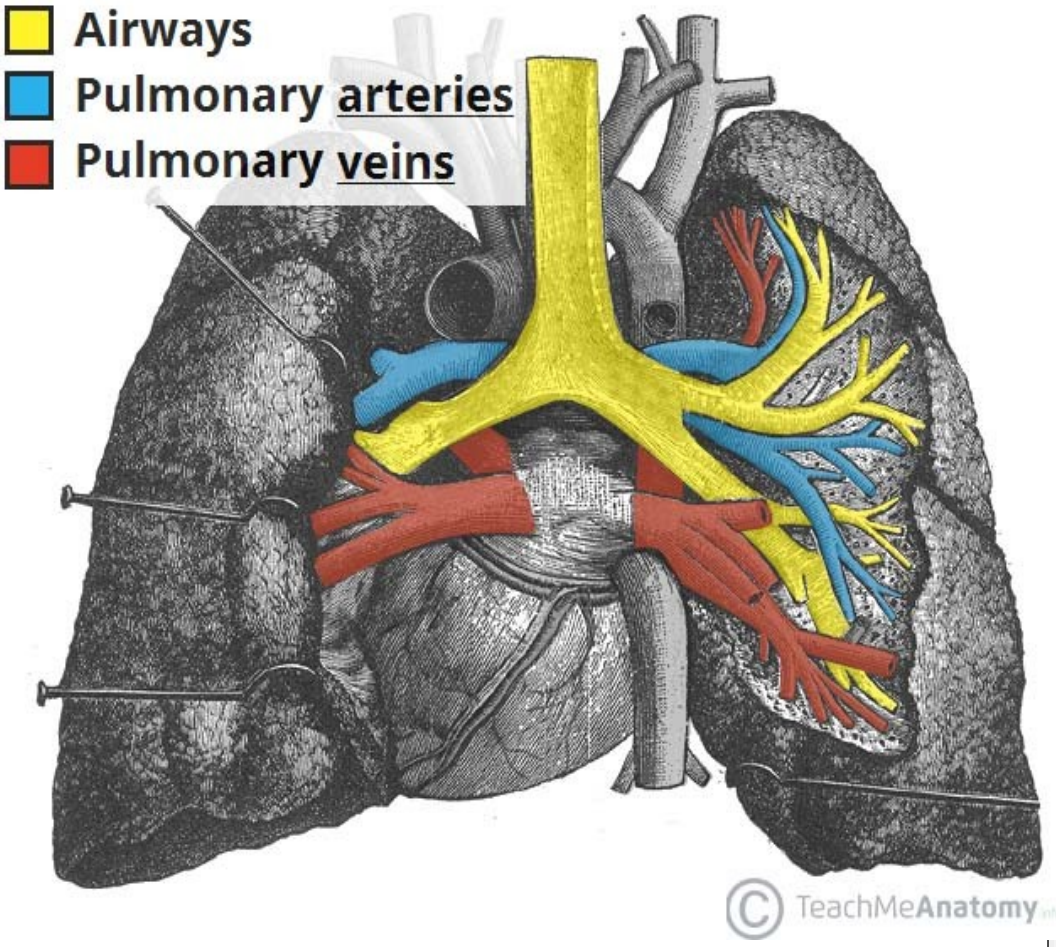
Left lung (lower lobe):
6. Superior
7. Medial basal
8. Anterior basal
9. Lateral basal
10. Posterior basal

Vasculature

- The lungs are supplied with deoxygenated blood by the paired **pulmonary arteries**. Once the blood has received oxygenation, it leaves the lungs via four **pulmonary veins** (two for each lung).

- **Blood supply: bronchial arteries** (which arise from the descending aorta)

- **venous drainage:**
  - The right bronchial vein drains into the azygos vein
  - the left drains into the hemiazygos vein.



Nerve supply

- The nerves of the lungs are derived from the **pulmonary plexuses**. They feature sympathetic, parasympathetic and visceral afferent fibres:
- **Parasympathetic**: Derived from the vagus nerve. They stimulate secretion from the bronchial glands, contraction of the bronchial smooth muscle, and vasodilation of the pulmonary vessels.
- **Sympathetic**: Derived from the sympathetic trunks. They stimulate relaxation of the bronchial smooth muscle, and vasoconstriction of the pulmonary vessels.
- **Visceral afferent**: Conduct pain impulses to the sensory ganglion of the vagus nerve.

Lymphatic Drainage Lungs

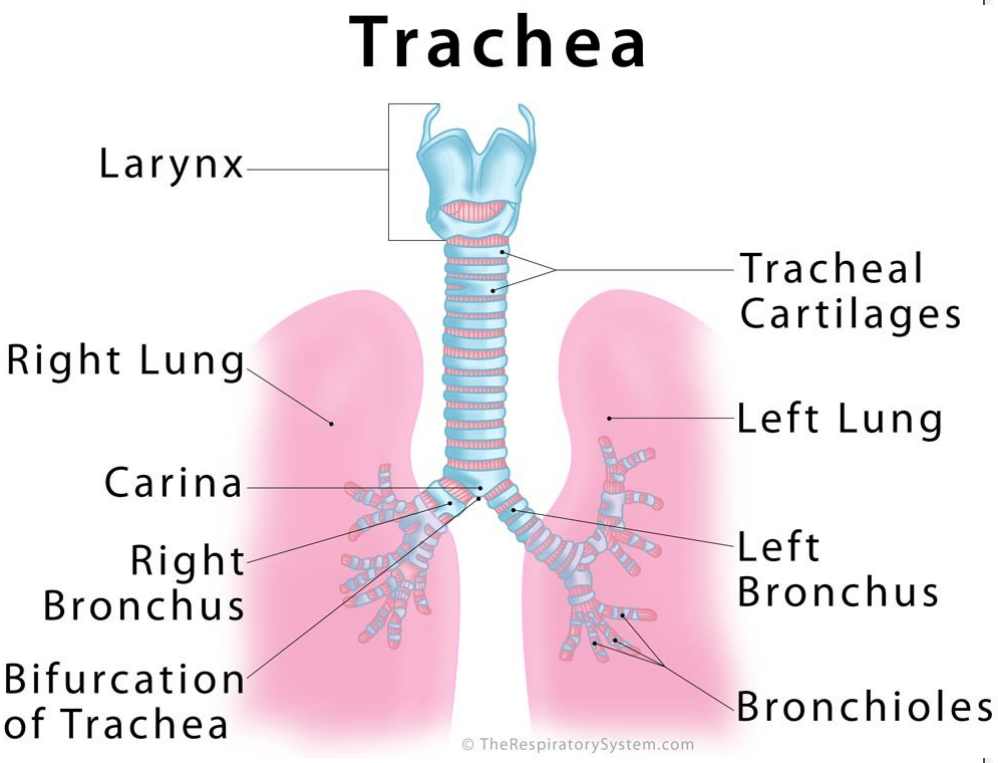
- Superficial and deep plexuses
- Superficial is immediately deep to visceral pleura
- Deep is formed deeply in lungs
- sup and deep plexuses drain into:
  - Bronchoulmonary(hilar) nodes (located in hilum)
  - ↓
  - Tracheobronchial nodes (located at bifurcation of trachea)
  - ↓
  - Bronchomediastinal nodes (right & left)
  - ↓
  - Bronchomediastinal trunks (right & left)
  - ↓
  - Right side → right lymphatic duct
  - Left side → thoracic duct

Trachea

- **Anatomical position:**
  - Begins at inferior border of cricoid cartilage
  - Bifurcates at sternal angle (T4)
  - the trachea is located anteriorly to the **oesophagus**
- **Structure:**
  - 16 to 20 incomplete hyaline cartilage rings that prevent trachea from collapsing
  - The free ends of these rings are supported by the **trachealis muscle**.
  - At the bifurcation of the primary bronchi, a ridge of cartilage called the **carina** runs between the openings of the two bronchi.
- Trachea May be compressed by aortic arch aneurysm, goiter and thyroid tumor

Neurovascular Supply

- The trachea receives sensory innervation from the **recurrent laryngeal nerve.**
- Arterial supply comes from the tracheal branches of the **inferior thyroid artery**, while venous drainage is via the brachiocephalic, azygos and accessory hemiazygos veins.



Bronchus

At sternal angleTrachea bifurcates into two bronchi:

Right bronchus

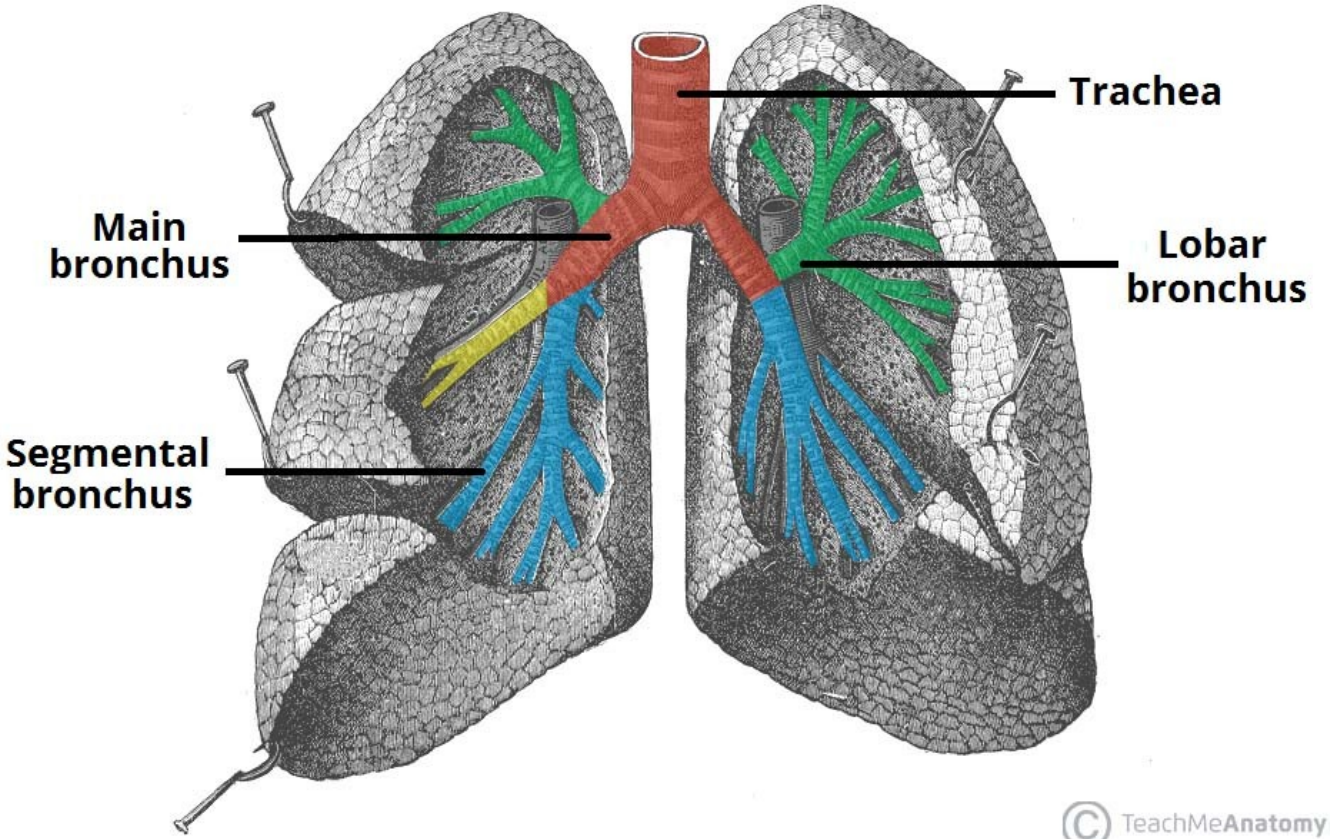
- It is shorter, wider, and more vertical
- Foreign body entering through trachea are mostly lodged into right bronchus.
- Runs under the arch of azygous vein.
- Further divided into three lobar bronchi and 10 segmental bronchi

Left bronchus

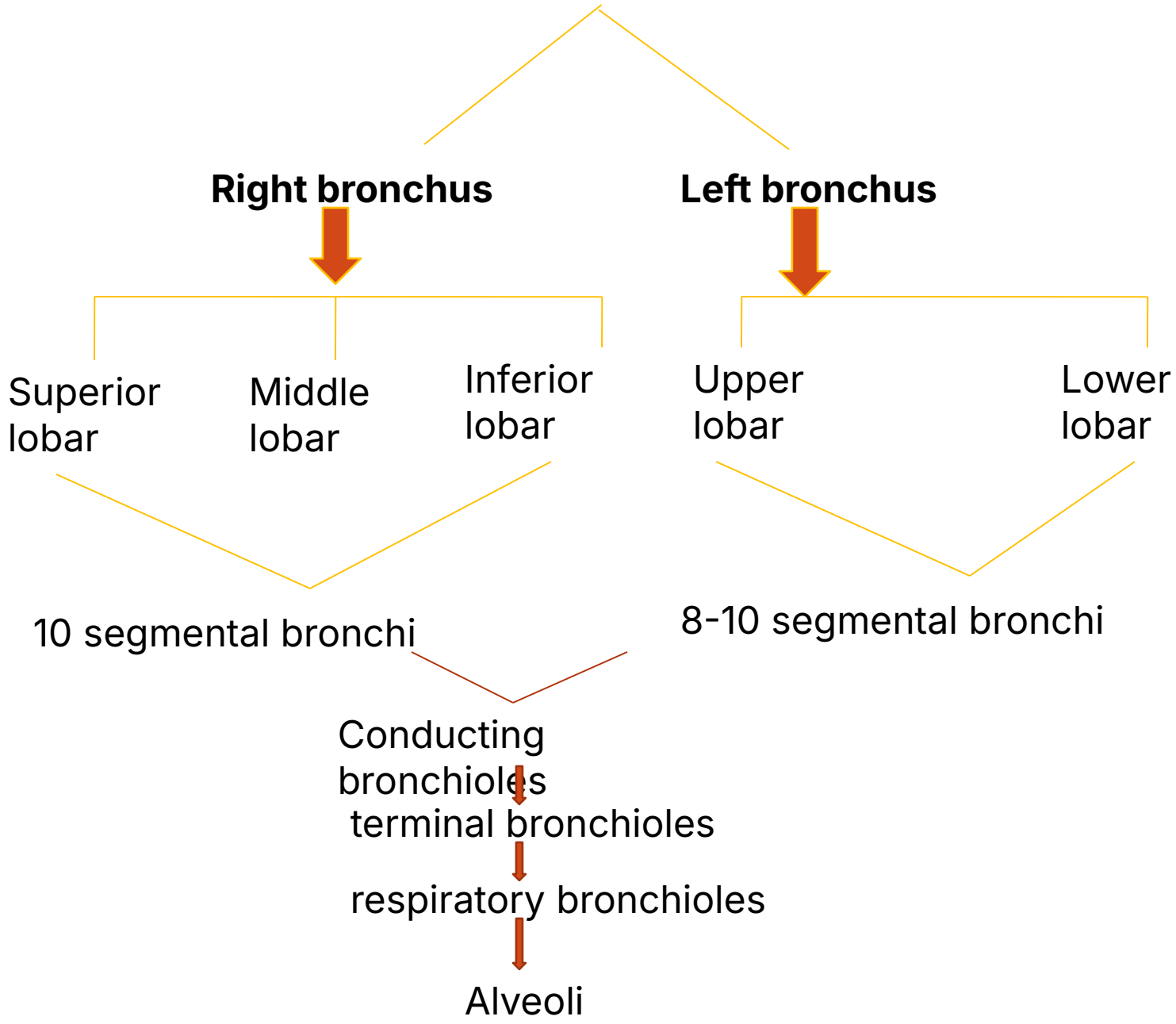
- It is longer , narrower and more horizontal
- It runs inferiorly to arch of aorta and left pulmonary artery.
- Crosses anteriorly to esophagus and thoracic aorta
- Further divided into two lobar bronchi and 8-10 segmental bronchi

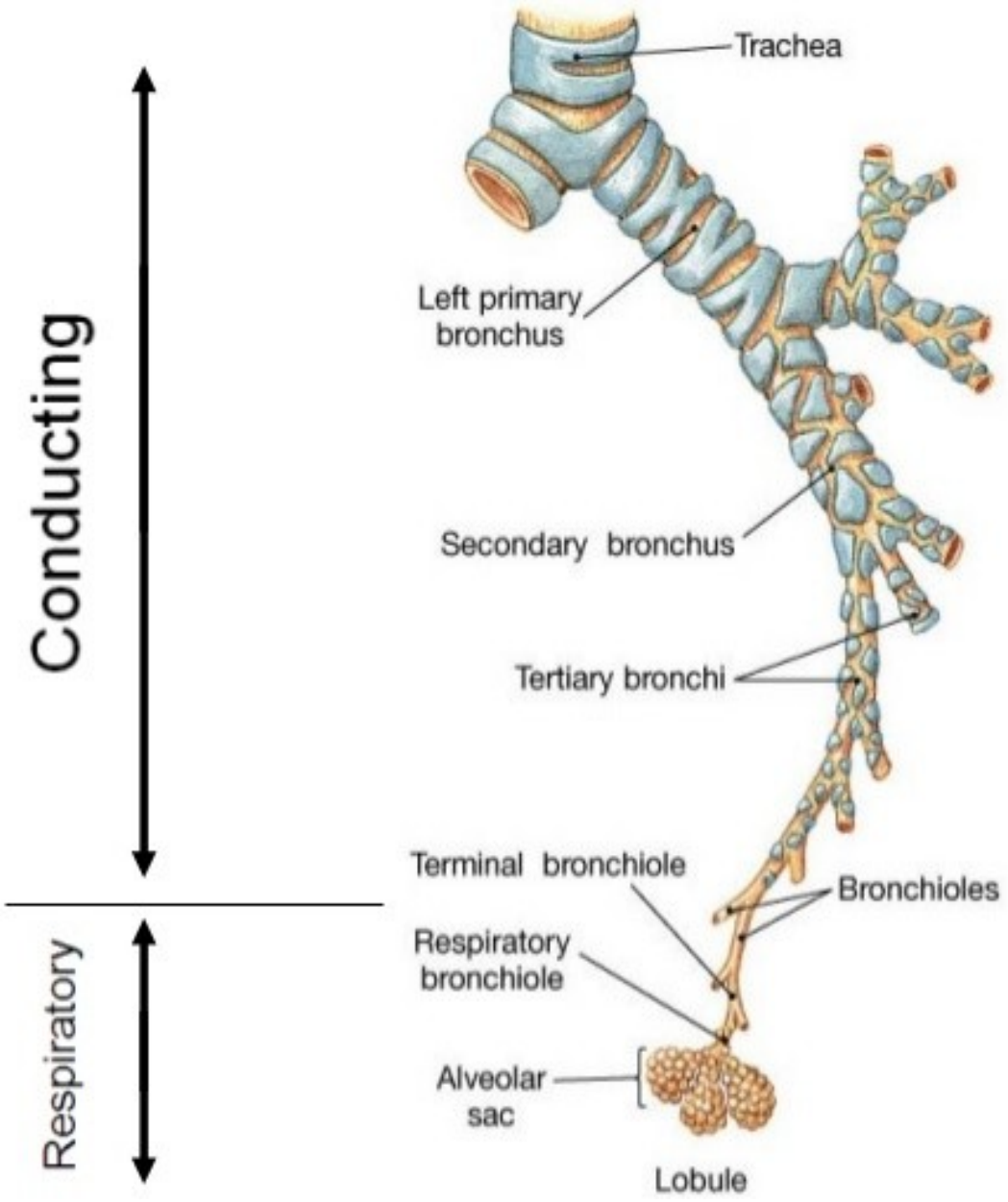
Neurovascular Supply:

- The bronchi derive innervation from pulmonary branches of the **vagus nerve** (CN X).
- Blood supply to the bronchi is from branches of the **bronchial arteries**
- veno



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TRACHEA
Right bronchus
Left bronchus
Superior lobar
Middle lobar
Inferior lobar
Upper lobar
Lower lobar
10 segmental bronchi
8-10 segmental bronchi
Conducting bronchioles
terminal bronchioles
respiratory bronchioles
Alveoli


Conducting
Respiratory
Trachea
Left primary bronchus
Secondary bronchus
Tertiary bronchi
Terminal bronchiole
Bronchioles
Respiratory bronchiole
Alveolar sac
Lobule

Airway

Conducting airway

- NASAL CAVITY
- NASOPHARYNX
- LARYNX
- TRACHEA
- BRONCHI
- BRONCHIOLES

Respiratory airway

- RESPIRATORY BRONCHIOLES
- ALVEOLAR DUCTS
- ALVEOLI

Thank you